

EIS study on the formation of solid electrolyte interface in Li-ion battery

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Abstract

Formation of solid electrolyte interface (SEI) on the surface of graphite in Li/graphite cells was studied using the electrochemical impedance spectroscopy (EIS) technique. Results show that EIS of the cells varies considerably with cell voltage. In particular, resistance (R_{sei}) of the SEI increases significantly with lithiation and decreases reversibly in the subsequent delithiation in a narrow voltage range between 0.15 and 0.04 V. According to the change of the R_{sei} during the first lithiation process, we might divide the formation of the SEI into two voltage regions: the first region generally occurs above 0.15 V and the second region below 0.15 V. The SEI formed in the first region is less conductive, and that formed in the second region is highly conductive. It is shown that ionic conductivity of the preliminarily formed SEI is greatly affected by the kind of solvents and salts of the electrolyte. We observed that the R_{sei} increases in the order of $\text{LiBF}_4 > \text{LiSO}_3\text{CF}_3 > \text{LiBOB} > \text{LiPF}_6$ for salt, and of $\text{NMP} > \text{EMC} > \text{MB}$ for solvent. Addition of vinylene carbonate into the electrolyte results in a significant increase of the R_{sei} .

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1. Introduction

Natural graphite has been a preferred anode material for the Li-ion batteries due to its high capacity, good cyclability and low potential versus lithium metal. However, fresh surface of the graphite has more or less catalytic effect on the electrochemical reduction of the electrolyte solvents, which reduces reversibility of the energy storage [1–3]. Meanwhile, solvent molecules in the electrolyte can cross through the fresh surface to intercalate into graphite and subsequently decompose between graphite planes, which can consequently exfoliate planar structure and decay capacity of the graphite [4–6]. It has been known that the above undesirable phenomena can be suppressed effectively by forming a solid electrolyte interface (SEI) on the surface of graphite. For successful Li-ion batteries, a preliminary SEI must be formed in the first cycle, especially during the first lithiation process (or called intercalation of Li^+ ions into graphite), and the

SEI morphology can be matured in the subsequent cycles. Therefore, the formation of the SEI has been considered to be an essential process for the fabrication of Li-ion batteries. Because the SEI is a surface layer that allows ions to move freely and isolates electron flow, and its main chemistry is the insoluble decomposed products of the electrolyte components, the formation of the SEI can be studied in situ by the electrochemical impedance spectroscopy (EIS) technique.

For a Li/graphite cell, EIS reflects a combined effect of the impedances of graphite electrode, electrolyte, and Li electrode. During lithiation and delithiation cycling, however, contributions of the electrolyte and lithium electrode to the EIS are believed to be invariant due to a result of the instant stripping and plating of lithium, which remains the surface of lithium constantly fresh. Therefore, any change in the EIS during the cycling process could be considered as the contribution of graphite. For this reason, in this work we have used two-electrode Li/graphite cells as the test object to study the formation of the SEI by recording EIS and comparing its change with the formation. We discuss the effect of solvents and salts of the electrolyte on ionic conductivity of the SEI.

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2. Experimental

Standard graphite anode of the Li-ion batteries, provided by SAFT America, Inc., was used to study the formation of the SEI. The graphite electrode (film) was punched into small discs with an effective electrode area of 1.27 cm^2 and dried at 110°C under vacuum for 16 h before use. Solvents of ethylene carbonate (EC, Grant Chemical), ethylmethyl carbonate (EMC, EM Industries), methyl butyrate (MB, Aldrich), γ -butyrolactone (BL, Aldrich) and 1-methyl-2-pyrrolidinone (NMP, Aldrich) were dried over $4 \text{ \AA}$ molecular sieves for a week and determined by Karl–Fisher titration to have lower than 20 ppm of water. Vinylene carbonate (VC, Aldrich), LiPF_6 , and LiBF_4 (both from Stella Chemifa Corp.) were used as received. Lithium bis(oxalate) borate (LiBOB, Chem. Metal) and LiSO_3CF_3 (Aldrich) were dried at 105°C under vacuum for 8 h prior to use. Using the solvents and salts above, a series of electrolytes with a 1.0 m (molality, i.e., mol salt per kilogram solvent) lithium salt dissolved in a mixed solvent was prepared in an argon-filled glove-box. Li/graphite button cells were assembled and filled with $100 \mu\text{l}$ of liquid electrolyte in the same glove-box.

A Solartron SI 1287 electrochemical interface and a SI 1260 impedance/gain-phase analyzer, controlled by CorrWare and Zplot softwares, were used for EIS measurement. The EIS were potentiostatically measured at a specific voltage with an ac oscillation of 10 mV amplitude over the frequencies from 100 kHz to 0.01 Hz. The voltage at which EIS was measured was achieved by cycling the cell at a constant current density of 0.1 mA/cm^2 . The collected EIS were fitted using ZView software. All tests were performed at 30°C , which was controlled by a Tenney Environmental Oven Series 942.

3. Results and discussion

3.1. EIS of the Li/graphite cell

EIS of a typical Li/graphite cell at various voltages are plotted in Fig. 1. In most of voltage range, the EIS are composed of two partially overlapped semicircles and a straight sloping line at low frequency end. Such a pattern of EIS can be fitted by an equivalent circuit as shown in Fig. 2 [7–11]. R_e is bulk resistance of the cell, which reflects a combined resistance of the electrolyte, separator and electrodes. R_{sei} and C_{sei} are resistance and capacitance of the SEI on the surfaces of two electrodes, which correspond to the semicircle at high frequencies. R_{ct} and C_{dl} are charge-transfer resistance and its relative double-layer capacitance, which correspond to the semicircle at medium frequencies. W is the Warburg impedance related to a combined effect of the diffusion of lithium ions on the electrode–electrolyte interfaces, which corresponds to the straight sloping line at low frequency end. The combination of charge-transfer impedance and Warburg impedance is called Faradic impedance, which reflects kinet-

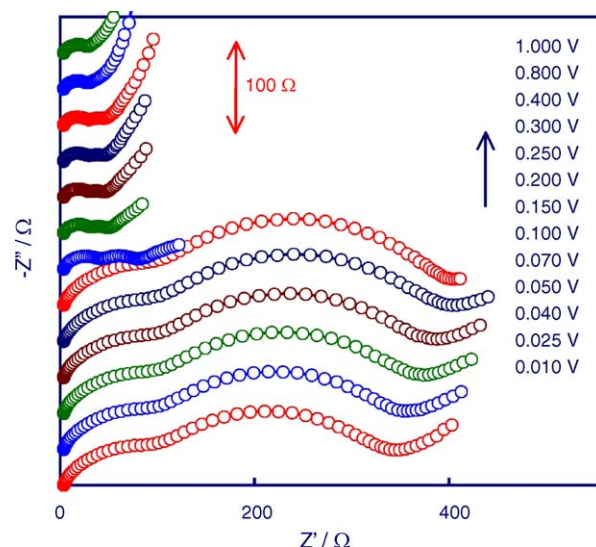


Fig. 1. EIS of the Li/graphite cell at various voltages, which were recorded during the first delithiation of a cell with 1.0 m LiPF_6 3:7 (wt.) EC/EMC electrolyte.

ics of the cell reactions and varies considerably with the state-of-lithiation of graphite [11,12]. Fig. 1 shows that the semicircle for charge-transfer process became unobservable as the voltage was higher than 0.8 V, in which graphite is fully delithiated and no other electrochemical reactions take place. In such cases, the R_{ct} is too high so that its semicircle cannot be formed. Based on the fact that formation and growth of the SEI is accompanied by change of the R_{sei} , we may use EIS as an in situ tool to study the formation of the SEI.

3.2. Understanding of the $R_{\text{sei}}-E$ correlation

The R_{sei} in the first and 10th cycles are plotted together with the differential capacity as a function of the cell voltage in Fig. 3. To understand $R_{\text{sei}}-E$ correlation of the first cycle,

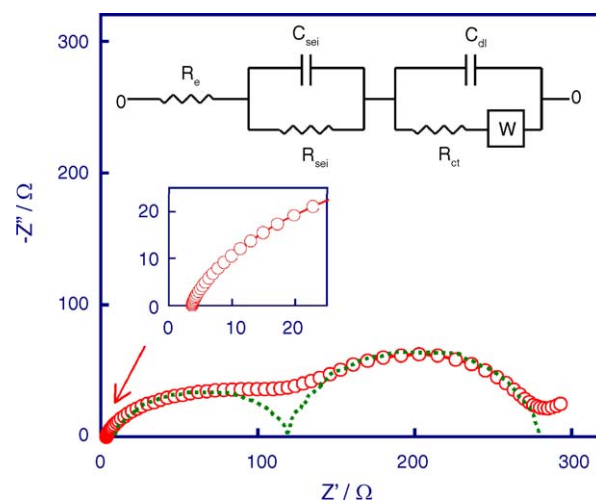


Fig. 2. Equivalent circuit for the EIS of Li/graphite cell. The shown EIS was recorded at 0.05 V.

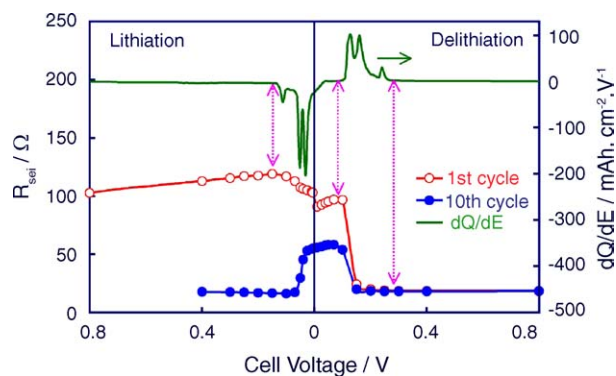


Fig. 3. Voltage dependence of the differential capacity and the R_{sei} of the Li/graphite cell, which was measured from a cell with 1.0 m LiPF₆ 3:7 (wt.) EC/EMC electrolyte.

we first analyze the one of the 10th cycle, at which the formation of the SEI is assumed to be complete. It is obvious that the R_{sei} increases significantly at ~ 0.07 V in the lithiation process and decreases reversibly to the original level at ~ 0.1 V in the delithiation process. The similar changes have been observed not only from other Li/graphite cells [11,13,14] but also from Li/cathode cells [15–17]. We contribute such changes to the expansion and extraction of graphite volume, which are caused by the lithiation and delithiation processes. It could be a general rule that expansion of the electrode volume results in an R_{sei} increase and extraction leads to an R_{sei} decrease.

Owing to the formation of the SEI, the $R_{\text{sei}}-E$ correlation in the first cycle becomes rather complicated. We might divide the $R_{\text{sei}}-E$ plot of the first lithiation into two regions. Above 0.15 V at which the lithiation process has not initiated (as indicated by differential capacities of near zero in Fig. 3), the R_{sei} increases slowly with decreasing of the cell voltage. This behavior is attributed to the formation of a preliminary SEI, which is relatively resistive [11]. Between 0.15 and 0.04 V during which the lithiation takes place (as indicated by three peaks of the differential capacities in Fig. 3), the R_{sei} declines quickly with decreasing of the cell voltage, being opposite to the observation in the 10th cycle. This result suggests that a highly conductive SEI must be formed from 0.15 to 0.04 V, which not only compensates the increase of the R_{sei} relating to the expansion of graphite volume but also results in an apparent decrease of the R_{sei} . Upon subsequent delithiation (as indicated by other three peaks of the differential capacities in Fig. 3), the R_{sei} starts to decrease dramatically around 0.1 V and remains invariant above 0.3 V at which the delithiation completes. Based on the above results, we conclude that the SEI is formed in two voltage regions: The first region is above 0.15 V and the second region is below 0.15 V. It should be noted that the voltage of the border of two regions might change slightly from cell to cell. This is because of a small difference in the electric polarization between different cells. Reliable voltage of the border should be at the onset of the first peak in the plot of differential capacity versus cell voltage of the lithiation process. In addition, Fig. 3 shows that at

Table 1

Ionic conductivity of the liquid electrolytes at 23 °C

Effect of solvents		Effect of salts	
1.0 m LiPF ₆ in	mS cm ⁻¹	1.0 m in 3:7 EC/EMC	mS cm ⁻¹
1:1 EC/EMC	8.3	LiPF ₆	8.2
3:3:4 EC/EMC/EMC	8.2	LiBF ₄	3.3
3:3:4 EC/EMC/MB	9.3	LiBOB ^a	5.6
3:3:4 EC/EMC/NMP	9.9	LiSO ₃ CF ₃	1.7

^a 0.8 m due to its limited solubility in the solvent.

fully lithiated state (around 0 V), the R_{sei} in the 10th cycle is much lower than in the first cycle. This is likely because morphology of the SEI is further matured in the subsequent cycles, which results in the R_{sei} decreasing.

It is important to understand the $R_{\text{sei}}-E$ correlation in the second voltage region (i.e., below 0.15 V in Fig. 3), in which the lithiation takes place. As discussed above, the R_{sei} in this region is affected by two opposite effects: (1) formation of the highly conductive SEI, which results in a decrease of the R_{sei} and (2) expansion of the graphite volume, which leads to an increase of the R_{sei} . Depending on the interaction of these two opposite effects, the R_{sei} may decrease, increase or remain unchanging with decreasing of the cell voltage.

3.3. Effect of solvents and salts on the R_{sei}

Effect of the electrolyte components on the R_{sei} of Li/graphite cells is studied by using different electrolytes whose composition and ionic conductivity at room temperature (23 °C) are summarized in Table 1. For study on the effect of solvents, a 1:1 (wt.) EC/EMC mixture was used as the baseline and a third solvent was added into the baseline. Fig. 4 compares the $R_{\text{sei}}-E$ plots of the Li/graphite cells with four different electrolytes. It is shown that at fully lithiated state (i.e., near 0 V), the R_{sei} increases in the order of MB > EMC > baseline > NMP. That is, addition of MB and EMC into the baseline electrolyte increases R_{sei} and addition of NMP reduces R_{sei} . This result cannot be explained by the ionic conductivity of the electrolytes since all these

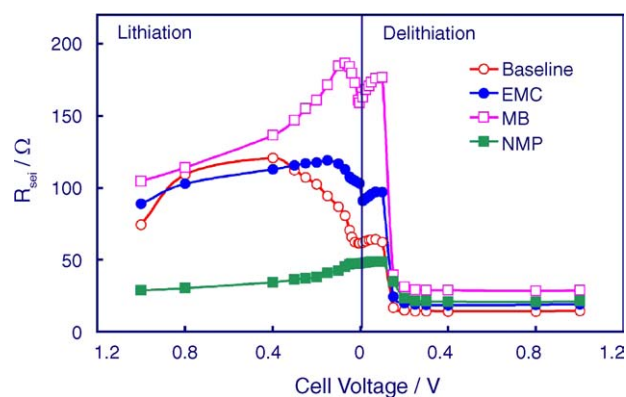


Fig. 4. Effect of solvents on the $R_{\text{sei}}-E$ plots, which were obtained from the first cycle of the Li/graphite cell. Composition and conductivity of the electrolytes are shown in Table 1.

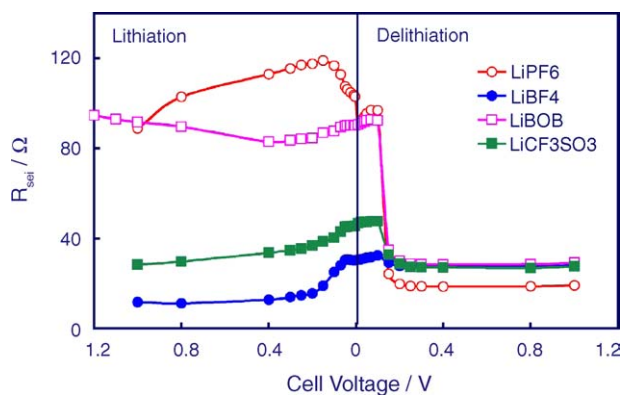


Fig. 5. Effect of salts on the $R_{\text{sei}}-E$ plots, which were obtained from the first cycle of the Li/graphite cell. Electrolyte was a solution prepared by dissolving 1.0 M lithium salt in a 3:7 (wt.) EC/EMC mixture.

electrolytes have the similar conductivity (Table 1). We consider that this observation can be attributed to the reductive reactivity of the solvents on the surface of graphite. It was known that at low potentials linear monoesters, such as MB, are more reductively reactive, which can result in the formation of a thicker and resistive SEI [18]. Therefore, the R_{sei} is increased significantly with addition of MB into the baseline electrolyte.

The effect of salts on the R_{sei} is studied by dissolving 1.0 M different lithium salt into a 3:7 (wt.) EC/EMC mixture. Fig. 5 shows the $R_{\text{sei}}-E$ plots of the first cycle of Li/graphite cells. It is shown that the R_{sei} at the fully lithiated state increases in the order of $\text{LiPF}_6 > \text{LiBOB} > \text{LiSO}_3\text{CF}_3 > \text{LiBF}_4$. This behavior also cannot be explained in terms of the ionic conductivity of the electrolytes (see Table 1). The electrolyte with LiPF_6 salt has the highest conductivity (8.2 mS cm^{-1} at 23°C), however, the SEI formed with it exhibits the highest R_{sei} . This is because LiPF_6 is most reactive to the electrolyte solvents. It has been reported [19] that the presence of electrolyte solvents can induce and accelerate decomposition of LiPF_6 , i.e., $\text{LiPF}_6 \rightarrow \text{LiF} + \text{PF}_5$. The resulting PF_5 not only reacts with the electrolyte solvents but also catalyzes polymerization of the solvents [20]. Therefore, the SEI formed with LiPF_6 could be thick and highly resistive. For the SEI formed with LiBOB , the R_{sei} decreases above 0.3 V with decreasing of the cell voltage. This exception can be attributed to the small amount of impurity in LiBOB , which has been identified to decompose around 1.8 V and participate into the SEI [21]. It is interesting to notice the difference of the $R_{\text{sei}}-E$ correlation below 0.15 V. Except for the one with LiPF_6 , all R_{sei} with other three salts are increased with decreasing of the cell voltage. This is because the R_{sei} of the initially formed SEI for these salts are relatively low and the effect of the expansion of graphite volume becomes predominant.

3.4. Effect of vinylene carbonate on the R_{sei}

It has been known that addition of small amount of vinylene carbonate (VC) into the electrolyte can significantly

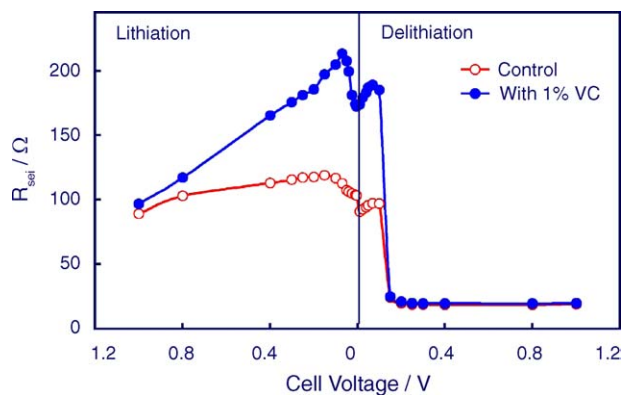


Fig. 6. Comparison of the $R_{\text{sei}}-E$ plots for the Li/graphite cells without and with vinylene carbonate, which were recorded during the first cycle. Control electrolyte was a 1.0 M LiPF_6 3:7 (wt.) EC/EMC solution.

enhance stability of the SEI since it participates into the formation of the SEI [22,23]. Therefore, VC has been extensively studied as the electrolyte additive of Li-ion batteries. In this work, the effect of VC on the R_{sei} is studied and displayed in Fig. 6. It can be seen that the R_{sei} is increased significantly by addition of VC into the electrolyte. This is because VC is polymerized electrochemically at higher potentials (before lithiation initiates) and the resulting polymer is combined into the SEI. At fully delithiated state, however, the R_{sei} of these two cells without and with VC are returned to the same level of about 20 Ω. The similar phenomena also are observed from Figs. 4 and 5. Based on the above results, we conclude that the electrolyte components, solvent, salt and additive, mainly affect the SEI of graphite at the fully lithiated state.

4. Conclusions

EIS of the Li/graphite cell is greatly affected by the cell voltage (or called state-of-lithiation). According to the $R_{\text{sei}}-E$ correlation, the formation of the SEI on the surface of graphite can be divided into two voltage regions. The first region is above 0.15 V, during which the formed SEI is less conductive. The second region is below 0.15 V, during which the formed SEI is highly conductive. For a formed cell, the R_{sei} reversibly increases and decreases with the lithiation and delithiation cycling, which are attributed to the expansion and extraction of graphite volume. In the second voltage region, the $R_{\text{sei}}-E$ correlation during the formation of the SEI is affected by two opposite effects: (1) formation of the highly conductive SEI, which results in a decrease of the R_{sei} and (2) expansion of the graphite volume, which leads to an increase of the R_{sei} . In addition, the R_{sei} during the first lithiation is very sensitive to the reactivity of the electrolyte components. In general, the R_{sei} increases with the reactivity of solvents and salts. On the other hand, the R_{sei} in the initial lithiation process is significantly increased by addition of small amount of vinylene carbonate into the electrolyte.

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